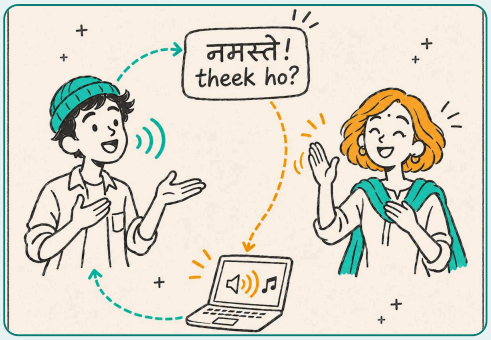
Samvaad संवाद

Two people. No interpreter in the room. Finally understanding each other.

A two-way bridge between deaf and hearing people. Live Hinglish captions in one direction, a natural spoken voice in the other. Software only: one laptop, one browser tab, zero hardware. Built for the Gnani AI Great Indian AI Internship Challenge.



samvaad-bridge.netlify.app | github.com/Hazz-Y/samvaad



The problem

Ananya is buying medicine for her grandmother. The pharmacist is rushed, the shop is loud, and Ananya is deaf. He rattles off the dosage while turning to the shelf and lipreading fails. **For medicine, a guess is not good enough.** Notes apps are slow, and the whole queue is watching.

> **Samvaad turns that laptop into a bridge.** His words become large live captions. Her taps, typed text and hand signs become a spoken voice. Both directions land in one conversation thread.

How a conversation flows



They speak naturally. Only speech leaves the laptop; silence is gated out.



You read big captions. Hinglish rendered the way you read it.

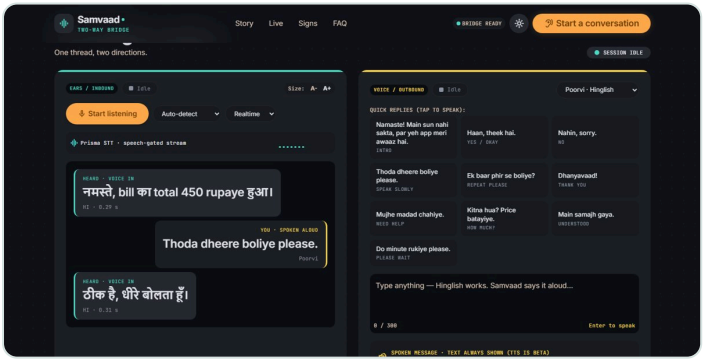


You tap, type, or sign. Quick phrases, free text, or 16 hand signs.

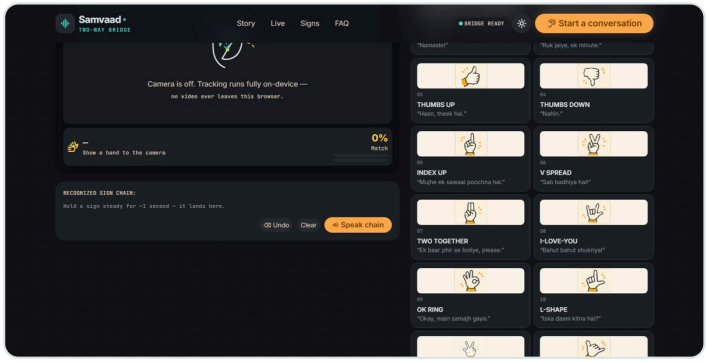


Samvaad speaks it aloud. Warm neural voice, text always shown too.

The product, running



One thread, two directions. Heard on the left, spoken back on the right. Exportable as a `.txt` record. Live API stats shown: STT processing time, language, TTS latency.



Sign studio. MediaPipe hand tracking (21 keypoints) plus a custom geometric classifier: 16 signs, temporal smoothing, live confidence meter. Runs fully in the browser.

Powered by Gnani

- **Prisma** realtime STT over WebSocket (`prisma-v3.0`, 16 kHz PCM)
- **codemix** output: Devanagari for Hindi, Latin for English words, as people actually read
- **Timbre** TTS (`timbre-v2.5/v3.0`), default voice Poorvi in Hinglish

Built secure and lean

- API key never reaches the browser; FastAPI proxy / Netlify Functions
- Hand tracking is 100% on-device; nothing uploaded
- Silence is never sent; repeat phrases hit a TTS cache (zero credits)
- Auto-fallback to chunked REST on Netlify without stopping the mic

Honest scope

- 16 signs, **not** full Indian Sign Language; labeled "limited vocabulary"
- Every reply shows as text before the voice plays
- Not a replacement for a certified interpreter in medical or legal settings

Two ways. One conversation.

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